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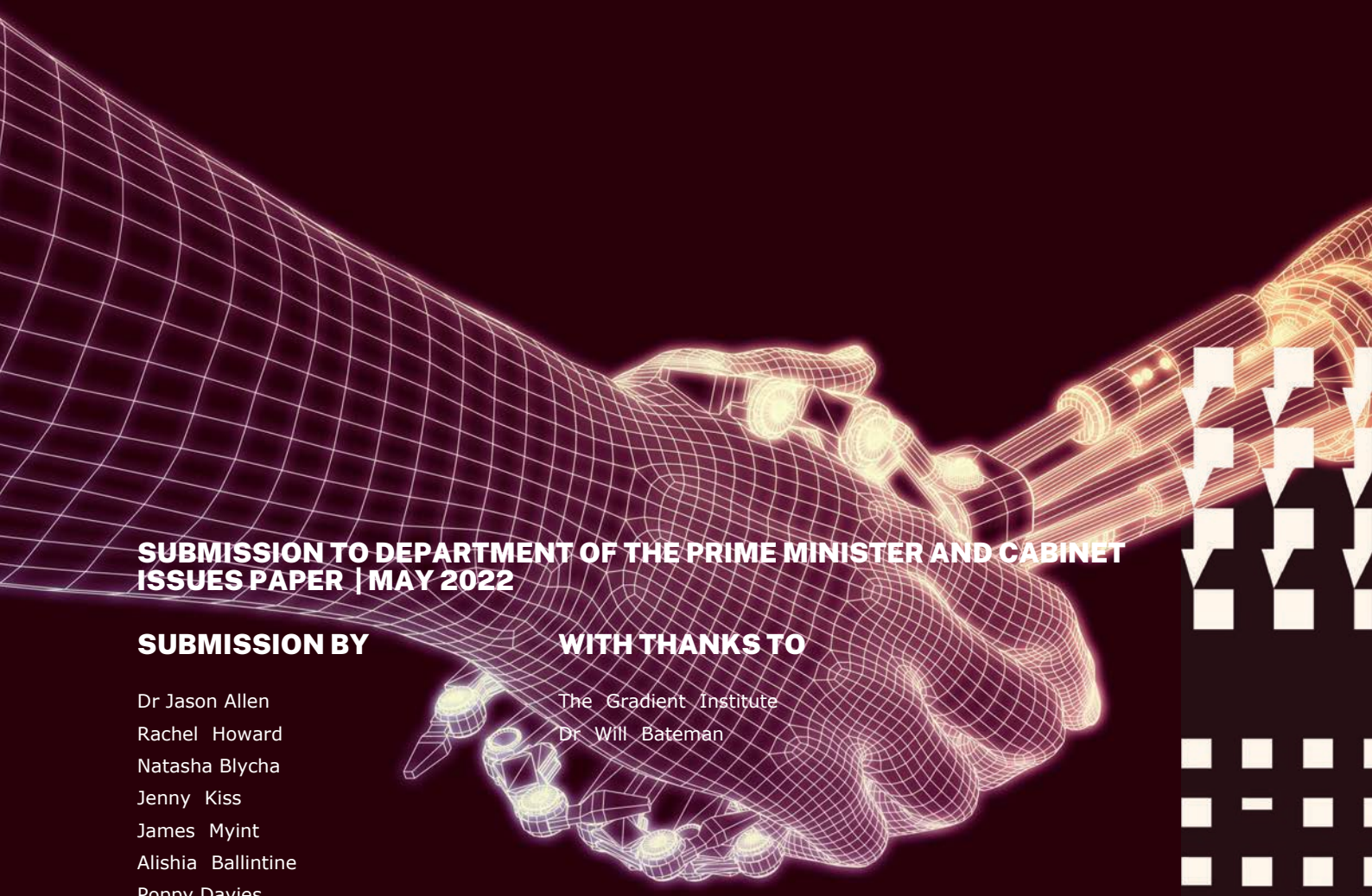
Positioning Australia as a leader in digital economy regulation

AI & AUTOMATED DECISION MAKING REGULATION

The time is ripe for decisive action in regulation to ensure Australia can drive beneficial innovation, enjoy a fair and functioning market and protect our citizens and consumers from economic and social harm.



SUBMISSION TO DEPARTMENT OF THE PRIME MINISTER AND CABINET ISSUES PAPER | MAY 2022

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WITH THANKS TO

The Gradient Institute
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|| Introduction: a robust regulatory framework can be a powerful catalyst for a healthy market

We welcome this Issues Paper and endorse the stated policy for Australia to become a global leader in digital economy regulation.

We believe that this aspiration is realistic, that action by the Australian Government is warranted, and that a regulatory framework can be a powerful catalyst for a robust and healthy market in AI and ADM. Indeed, rather than less (or no) regulation, responsible innovators and businesses welcome the certainty provided by appropriate regulatory guidelines.

In this submission, we set out some high-level responses to the Issues Paper from a commercial and practical legal perspective.

As we read it, the substance of the Issues Paper boils down to three essential questions:

- Why regulate AI & ADM—what is at stake?
- What is the role of the Australian Government in the regulation of AI & ADM?
- How should the Australian Government perform this role to protect the values at stake?

These questions frame the Issues Paper and underlie the ten questions it sets out. We have set out our responses under these headings rather than in an itemised fashion.

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|| Defining the terms of the enquiry

At the outset, it is convenient to outline our understanding of 'Artificial Intelligence' (AI) and 'Automated Decision Making' (ADM). AI and ADM are clearly overlapping categories; at their broadest, both can consist of (i) inputs, (ii) pre-defined objectives, (iii) techniques and approaches to achieve those objectives, and (iv) outputs.[1]

The borders of AI are notoriously difficult to define,[2] but the term is usually used to refer to 'Machine Learning' (ML) technologies. As discussed in Section 3, the key issues presented by these systems are agentivity (i.e., outputs that mimic human 'action'), opacity, and unpredictability, commonly referred to as the 'black box' problem.[3] We take a broad definition of 'ADM' to include any system (whether rule-based or data-driven) that automates some part of the process of decision-making.

The Issues Paper sets out some crucial issues of AI and ADM and administrative law principles. In administrative law, 'decision-making' (for example, by a Minister) has special importance.[4] We, however, do not define AI and ADM by reference to the private or public context of application. For example, algorithmic trading systems are of systemic importance in financial markets today, and their proper functioning is a matter that touches the public interest.[5]

The blurring of 'public' and 'private' in the mode of government service delivery and the importance of government contracting as an overarching consideration must also be kept in mind. It is thus essential to include 'private sector ADM' and 'public sector AI' in scope despite any labels of convenience.

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[1] See Teresa Rodriguez de las Heras Ballel, 'Guiding Principles for Automated Decision-Making in the EU' (European Law Institute Innovation Paper May 2022). https://europeanlawinstitute.eu/fileadmin/user_upload/p_eli/Publications/ELI_Innovation_Paper_on_Guiding_Principles_for_ADM_in_the_EU.pdf, 8.

[22] See generally discussion in Charles Kerrigan (ed), *Artificial Intelligence: Law and Regulation* (Edward Elgar 2022), [1.028].

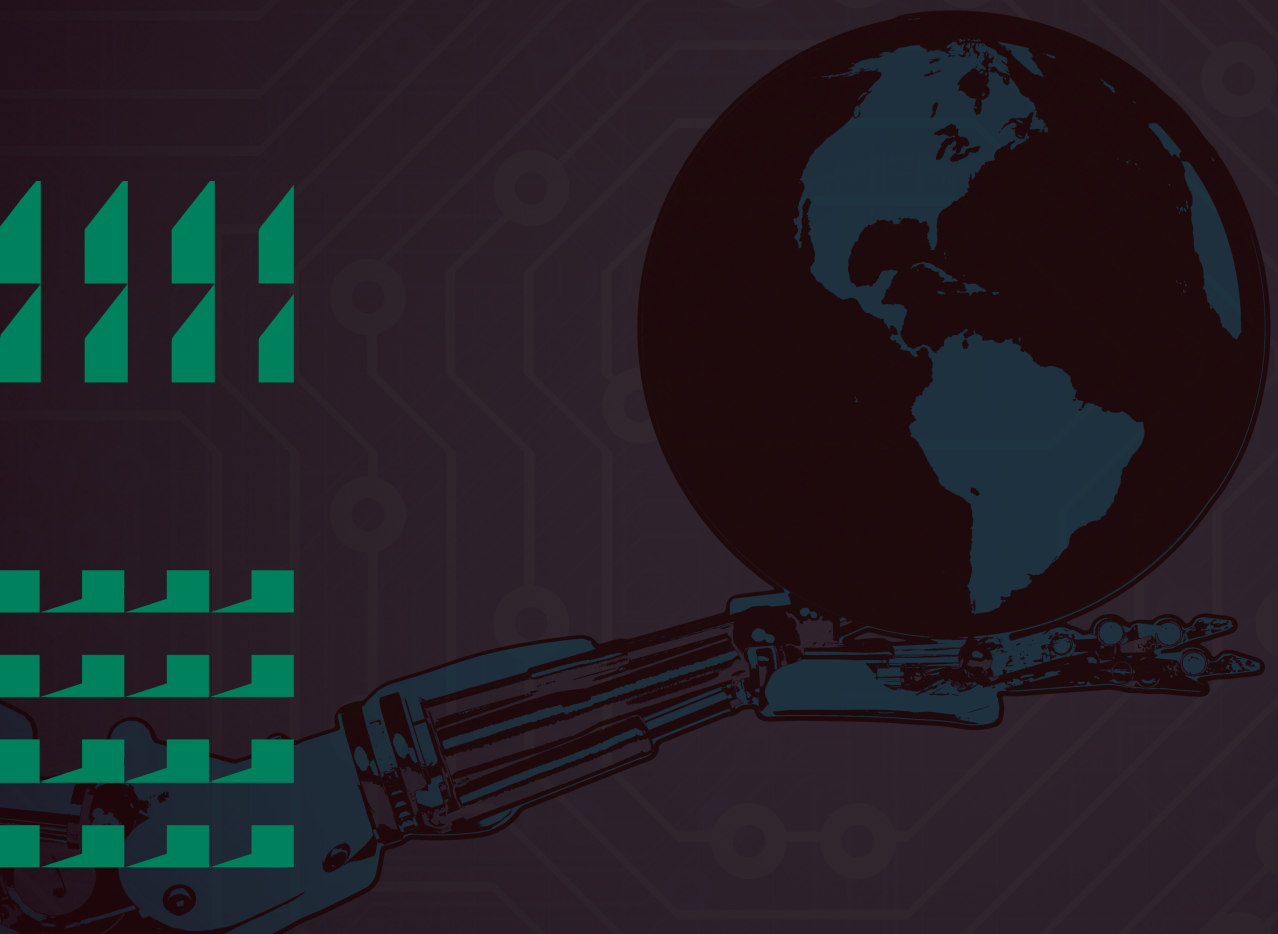
[3] That is, because these algorithms are data-driven rather than rule-driven, their outputs can never be completely predicted from their inputs.

[4] See J.G. Allen, 'Agency and Liability' in Charles Kerrigan (ed), *Artificial Intelligence: Law and Regulation* (Edward Elgar 2022), [9.039] and following.

[5] See US Securities and Exchange Commission, 'Staff Report on Algorithmic Trading on US Markets' (5 August 2020), https://www.sec.gov/files/Algo_Trading_Report_2020.pdf.

A BALANCING ACT

1. Timeliness
2. Inevitability
3. Social calculus
4. Principles based leadership
5. Facilitative and prudent regulation
6. Multi-level governance
7. Regional innovation and cooperation
8. Regulatory tools



|| Executive summary and recommendations

1. **Timeliness**: The time is ripe for decisive action in the regulation of AI and ADM with recent rapid developments in this technology. Accordingly, this regulation could be a powerful catalyst for beneficial market activity.
2. **Inevitability**: There are two inevitabilities. First, the arrow of technological progress will only move forward. Second, whether by organic local development of AI and ADM, direct importation of AI and ADM technologies into Australia or uptake in AI and ADM by Australia's trading partners which will mean the impact of AI and ADM flow to Australia, such technologies will arrive (and have already arrived) in Australia.
3. **Social calculus**: The cost-benefit proposition of AI and ADM is vast and rapid. There is the potential for mass productivity gains in terms of automation, but conversely, there are significant risks. The most obvious risk is job loss amongst the most vulnerable workforces (and certain professional services), and broader social inequality and reduction of real wages.
4. **Principles based leadership**: Leadership in digital economy regulation will not be achieved by 'slashing red tape' but by fostering fair and functioning markets (in the private sector context) and by the robust protection of administrative law values and principles (in the public sector context).
5. **Facilitative and prudent regulation**: Regulatory guide rails should establish a well-functioning, healthy market and avoid some of the centralising tendencies that we have seen in the last decade—mindful of the role of large technology platforms in the development and commercialisation of AI and ADM.
6. **Multi-level governance**: Industry, civil society, and international bodies all play a vital role in regulating AI and ADM. The Australian Government can draw on and reinforce efforts at various levels by providing a solid national framework that reflects emerging principles and best practices.
7. **Regional innovation and cooperation**: As an innovative jurisdiction, and a liberal democracy in the APAC region, Australia should see itself in a relationship of cooperation and coordination, not competition with other countries.

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8. **Regulatory tools**: Several regulatory levers are available, and the best suite (from time to time) should be chosen for deployment in any given context. These include at least the following elements:
   a. The establishment by legislative mandate of an Australian Sovereign Data Trust - the National Data Trust (**NDT**). The NDT would have legally mandated rights to de-identified data from large technology companies and public authorities. These would be pooled together with best practice de-identified and curated voluntary data contributions from, for example, universities, schools, churches, sporting groups and the private sector.
   b. Transparency and disclosure (for example, a publicly searchable AI and ADM register).
   c. Intelligent 'human in the loop' design that is sensitive to the relevant context and ensures the meaningful application of human intelligence in a complex system that includes technical and human components.
   d. Fostering a robust liability regime for AI and ADM in private law (for example, clear rules for fault-based liability in tort, no-fault liability, and possibly mandatory insurance akin to motor vehicle accident insurance).
   e. Ensuring clear accountability and liability rules for AI and ADM in administrative law (for example, ensuring that Ministers are accountable for decisions based on algorithmic systems).
   f. Dynamic supervision and oversight of applications by responsible public authorities.
   g. Robust competition and consumer protection frameworks to ensure a healthy, functioning market.
   h. Use by regulators of technology (i.e., 'RegTech', including AI and ADM technologies) to supervise applications in real time.
   i. Precautionary 'hard limits' to protect against systemic risks and risks of irreversible harms.

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// 1. Why should the Australian Government regulate AI and ADM?

'Innovation' *per se* is not a relevant policy goal for the Australian Government; the goal should be *beneficial* innovation for all Australians. If AI and ADM are desirable for a free, prosperous, and sustainable future, regulation must maximise such benefit whilst mitigating foreseeable harms. We cannot just impose ethical and moral standards reactively when things go wrong; regulation should be built around a coherent vision for AI and ADM in Australian society.

This requires an innovative regulatory approach that harnesses the power of a free and fair market, technology itself (ie, 'RegTech'), and the full suite of conventional regulatory levers.

The regulation of AI and ADM is essential to balance two policy goals that are often cast in opposition but are, in our view, mutually reinforcing:

- **Driving beneficial innovation** in the development and application of promising, general-purpose technologies.
- **Protecting** citizens, consumers, and systems from foreseen and foreseeable harms.

These policy goals are mutually reinforcing because, in the private sector at least, they are both most likely to be achieved through the creation and maintenance of a healthy and fair market. In some cases, this may include a more robust precautionary approach to 'known unknowns' and even 'unknown unknowns,' too.[6]

1.1 Responsible and effective regulation helps to drive beneficial innovation

The goal is to provide an effective framework for innovation to flourish in a way that does not cause harm. As recent experience in Australian Government service delivery has shown, public trust is essential for the sustainable adoption of emerging technologies.

Global expert opinion suggests *both* that AI and ADM have the potential to transform society positively, and that AI and ADM present significant risks to

[6] The reminder that 'absence of evidence isn't evidence of absence' is generally attributed to Martin Rees, O.M., Astronomer Royal, former master of Trinity College, Cambridge, and co-founder of the Centre for Existential Risk at the University of Cambridge with workstreams on *inter alia* AI.

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human societies.[7] Because AI and ADM are general-purpose technologies, the potential benefits and risks are context-specific. It would not be appropriate to attempt to capture them systematically in this submission. Generally, however, potential benefits of AI and ADM include:[8]

- Promoting speed and efficiency in decision-making
- Reducing costs in transactions
- Reducing error and human biases
- Assisting the solution of complex problems
- Monitoring and controlling increasing volumes of data

Potential harms from AI and ADM include:

- Exacerbating inequality and driving down relative wages through an excessive focus on simple automation[9]
- Disruption to democratic process and political interference
- Profiling of children to deliver content feeds which are harmful or inappropriate[9]
- Perpetuating privilege and bias embedded in data training sets
- Causing humans to lose control over crucial aspects of their lives
- Mass automated surveillance and data abuse, potentially leading to dystopian outcomes such as a citizen rating system
- Coercive nudge techniques which lead to loss of privacy or economic harm[10]
- Opaque decision-making that is not understood by the (nominal) decision-maker or its human users

[7] See Janna Anderson, Lee Rainie and Shawnee Cohn, 'Artificial Intelligence and the Future of Humans' (Pew Research Centre, December 2018), PI_2018.12.10_future-of-ai_FINAL1.pdf).
[8] This list has been compiled from various sources. See, e.g., Gradient Institute, 'Derisking Automated Decisions' (16 March 2022), https://gradientinstitute.org/wp-content/uploads/2022/03/gradient_minderoo_report.pdf.
[9] See Daron Acemoglu and Pascual Restrepo, 'Tasks, Automation, and the Rise in US Wage Inequality' (NBER Working Paper 28920, June 2021), https://www.nber.org/papers/w28920; Daron Acemoglu and Pascual Restrepo, Automation and New Tasks: How Technology Displaces and Reinstates Labor' (NBER Working Paper 25684. March 2019), https://www.nber.org/papers/w25684.
[10] See, e.g., the Age Appropriate Design: A Code of Practice for Online Services promulgated by the UK Information Commissioner's Office: https://ico.org.uk/for-organisations/guide-to-data-protection/ico-codes-of-practice/age-appropriate-design-a-code-of-practice-for-online-services/.

- Unforeseen and unforeseeable systemic (e.g., financial system or environmental) risks
- Dependence on AI and ADM for critical functions with no redundant (human) systems
- Doomsday risks such as weaponised AI and ADM (in the form of ICT disruption, control of physical infrastructure, and weaponised drones)

The consensus, even among those sanguine about the ability of AI and ADM to improve human futures, is that our reliance on technological systems will only go well if close attention is paid to how AI and ADM are engineered, distributed, and updated.

In our view, regulation plays an integral role by fostering functional markets within responsible guide rails. Sections 2 and 3 outline what that regulation might entail.

1.2 Additional consideration in the public sector

As the Issues Paper rightly states, there are some additional, nuanced considerations in the governmental use of AI and ADM. Administrative law protects core values such as the accountability of public officials (and others wielding 'public power'[11]) and the rule of law in its procedural and substantive aspects (e.g., the notion that laws must be understandable and certain to allow subjects to plan their lives[12]).

In our view, certain categories of public decision-making may not be amenable to certain AI and ADM tools at present — or at all. Although efficiency gains are important (for example, under the maxim that 'justice delayed is justice denied'), fundamentally, exercises of public power must remain comprehensible by the humans wielding it. Further, it must be explainable to those subject to it. In our view, this constrains the circumstances in which public agencies and those employed or contracted by them can deploy AI systems that are opaque or unpredictable, at least without a very robust regulatory framework. 'Computer says no' situations must be avoided, as must scenarios in which the logic of a

[11] See J.G. Allen, *Non-Statutory Executive Powers and Judicial Review* (Cambridge University Press 2022), Ch 12.
[12] See, e.g., Jeremy Waldron, 'The Concept and the Rule of Law' (2008) 43(1) *Georgia Law Review* 1.

procedure (embedded in a technical system) operates in a perverse way with no human able to hit the 'kill-switch'.[13]

The public sector is probably the first context in which the Australian Government should seek to regulate AI and ADM to ensure that automation adds value (e.g., efficiency) without than detracting from Australia's status as a liberal democratic state under the rule of law. Our homegrown experience and overseas examples of AI and ADM, for example, augmented sentencing decisions, provide some textbook tales of caution.[14]

In our view, the suitability for any decision to be automated depends on the type of decision to be made. For example, consider an application for a local government permit as of right against a checklist of criteria which are either 'fulfilled' or 'not fulfilled'. This assessment might be suitable for automation with minimal oversight (e.g., *ex post* on the complaint of the citizen). At the other end of the spectrum, discretionary decision-making by a judge or executive official that requires weighing or balancing competing interests presents a 'red light' scenario.

We would stress, however, that AI and ADM systems could be designed for legitimate application in any context. The touchstone question is whether the system is *augmenting* human intelligence—empowering humans to make better decisions—or *displacing* human intelligence in the search for speed or efficiency.

1.3 The role of large technology companies and training data

Another significant matter that bears noting relates to the current state of dominance by several large technology businesses that operate globally.[15] These businesses collect and own much of the data necessary to 'train' AI systems; the nature of those datasets directly informs the quality of an AI system.

[13] This kind of scenario is often associated with the literature of Franz Kafka, particularly his depiction of bizarre and impersonal administrative situations where the individual feels powerless to understand or control what is happening.

[14] In particular, the reproduction/amplification of racial bias in sentencing: see, e.g., Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, 'Machine Bias' (*ProPublica*, 23 May 2016), https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing; Will Douglas Heaven, 'Predictive policing algorithms are racist. They need to be dismantled' (MIT Technology Review, 17 July 2020), https://www.technologyreview.com/2020/07/17/1005396/predictive-policing-algorithms-racist-dismantled-machine-learning-bias-criminal-justice/.

[15] See Meredith Whittaker, 'The Steep Cost of Capture' (2021) 28(6) *ACM Interactions* 50.

This fact has various implications which we cannot explore fully here (but we would welcome further discussion), not least that even home-grown systems are likely to be trained on such datasets. However, the point is important because at least into the foreseeable future, Australia is (and is likely to remain) an importer of AI and ADM technology.[16]

AI development has been characterised as an arms race between China and the United States.[17] Thus, the political economy of AI innovation is likely to become more geopolitically sensitive in the future.

Further, Australia's regulatory challenge occurs amidst the increasing scrutiny by governments of dominant technology companies globally.

We emphasise that regulation of AI and ADM is an issue of national significance as AI and ADM tools become more important to the functioning of our economy and public administration. For example, if automation is a cause of job loss and Australia is likely to remain a net importer of automation technologies, regulation of those technologies could be cast as a national strategic priority.

[16] See, e.g., https://www.theguardian.com/business/2021/feb/06/is-big-tech-now-just-too-big-to-stomach.
[17] See, e.g., https://www.ft.com/content/21eb5996-89a3-11e8-bf9e-8771d5404543.

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// 2. What is the Australian Government's role in regulating AI and ADM?

Regulation of AI and ADM should primarily aim to create the framework for free and fair markets, healthy competition and the avoidance of monopolies, and robust avoidance of negative externalities and other textbook market failures. A healthy market is the primary way to promote the development of worthwhile products and services that improve consumers' and citizens' lives. Australia's national interests and the interests of Australian citizens must be protected if they are to prevail in a market dominated by a small number of very large businesses.

This being the proper focus of regulation, we suggest a subtle but important frameshift. Questions 1-6 of the Issues Paper suggest a 'slashing red tape' narrative. AI and ADM is a global social, economic, and (increasingly) strategic issue. Proactive policies that purposefully pursue the common good of Australian citizens and the national interest is better than taking a *laissez faire* approach and reacting retrospectively.

This does not require the Australian Government to engage in strategic planning of the digital economy—instead, to set parameters within which market and other social forces can unfold. This leads to a nuanced discussion around the 'What' and the 'How' of AI and ADM regulation.

2.1 Understanding 'regulation' in its broader context

Regulation has traditionally been viewed as 'command and control', i.e., the promulgation of rules by government and mechanisms of enactment, including monitoring and enforcement.[18] Such regulation has existed on a spectrum, from minimum to strong precautionary regulation. Given the dynamic and evolving nature of AI and ADM, a regulatory system that is flexible enough to accommodate this evolution yet prioritises public safety and minimises harm is paramount. The Australian Government's role should be agile and iterative—a coordinator, fostering national and international alignment and engagement across the ecosystem.

Australian Government coordination at the intersect of the public and private domain, vertically amongst government departments and horizontally across

[18] Julia Black, 'Critical Reflections on Regulation' (2002) 27 *Australian Journal of Legal Philosophy* 11.

different countries, disciplines, and technological domains is critical.[19] The notion that Australia is in 'competition' with other jurisdictions to regulate (or deregulate) AI and ADM[20] should be reframed. As a mid-sized but highly innovative jurisdiction, and a liberal democracy under the rule of law in the APAC region, Australia should rightly see itself in a relationship of cooperation and coordination, not competition.

Here, we would draw attention to the notion of multilevel stakeholder engagement. This approach allows for a broad and diverse spectrum of bodies that can respond to the challenges of automation in different ways and at various levels of power and influence.[21] The effective regulation of AI and ADM is unlikely to proceed as a matter of federal law and regulation, or indeed national law within any given country; rather, it will require coordinated initiatives at different levels:

- Within the **private sector** (for example, voluntary codes of conduct within specific industries such as the Australian Federal Chamber of Automotive Industries Voluntary Code of Conduct for Automotive Data and Privacy Protection[22]).
- Within **civil society** (for example, through the academy, professions, and third sector such as the European Law Institute Guiding Principles for Automated Decision-Making in the EU[23]).
- Within the Australian constitutional framework, initiatives at **local and state/territory levels** (for example, the Helsinki municipal AI register in Finland[24]).
- Internationally, at the **non-governmental level** (for example, NGO and international standards initiatives such as the International Standards

[19] See World Economic Forum, 'Global Technology Governance: A Multistakeholder Approach', (WEF White Paper, October 2019), https://www3.weforum.org/docs/WEF_Global_Technology_Governance.pdf.
[20] See the Issues Paper page 10.
[21] See World Economic Forum, 'Agile Governance: Reimagining Policy-making in the Fourth Industrial Revolution', (WEF White Paper, January 2018), https://www3.weforum.org/docs/WEF_Agile_Governance_Reimagining_Policy-making_4IR_report.pdf.
[22] See Federal Chamber of Automotive Industries, 'Voluntary Code of Conduct for Automotive Data and Privacy Protection', https://www.fcai.com.au/news/codes-of-practice/view/publication/172.
[23] Teresa Rodriguez de las Heras Ballel, 'Guiding Principles for Automated Decision-Making in the EU' (European Law Institute Innovation Paper May 2022). https://europeanlawinstitute.eu/fileadmin/user_upload/p_eli/Publications/ELI_Innovation_Paper_on_Guiding_Principles_for_ADM_in_the_EU.pdf.
[24] https://ai.hel.fi/en/ai-register/.

Organisation ISO/IEC AWI TS 6254 Objectives and approaches for explainability of ML models and AI systems[25]).

- Internationally at the **inter-governmental level** (for example, the OECD Principles on Artificial Intelligence[26] or the MOU between Australia and Singapore on AI under the Singapore Australia Free Trade Agreement[27]).

In our view, the Australian Government's role is to coordinate efforts and lead in developing and promulgating principles and best practices that tie all these efforts and initiatives together, resulting in a coherent whole at the national level.

There is a proliferation of documents and instruments with provisions relating to AI and ADM at present. As discussed below, there is an increasing degree of consensus internationally around several principles and best practices.

Mindful (i) of the multi-level nature of AI and ADM governance, (ii) of Australia's position in the region and the world, and (iii) of Australia's constitutional arrangements, we think that the time is ripe for leadership in crystallising some of these principles and best practices into a robust legal regime. In particular, we think it is important for Australia to align its approach with strategic and trading partners that share a similar commitment to the rule of law; specifically, Commonwealth jurisdictions such as the UK and Singapore, the US, and the European Union.

[25] https://www.iso.org/standard/82148.html?browse=tc.
[26] https://www.oecd.org/digital/artificial-intelligence/.
[27] https://www.dfat.gov.au/sites/default/files/australia-singapore-mou-artificial-intelligence.pdf.

// 3. How could the Australian Government perform its role in the regulation of AI and ADM?

The Australian Government should consider a diverse and context-specific toolbox of regulatory approaches and mechanisms. For example, different concerns arise in the context of (i) rule-based algorithmic trading in the cryptoasset markets, (ii) natural language processing in government service delivery, or (iii) the use of predictive data-driven algorithms in asylum seeker applications.

At a broad level, regulatory tools might include:

- The establishment by legislative mandate of an Australian Sovereign Data Trust - the National Data Trust. The NDT should have legally mandated rights to some reasonable proportion of de-identified data from large technology companies, as well as public authorities to be pooled with best practice de-identified and curated voluntary data contributions.

  In order to best train optimum and democratically accessible AI for the Australian public, we need to responsibly curate, control and share a national public utility style database par excellence. A public utility of this nature would generate positive network externalities and is highly likely to encourage other nations to similarly create sovereign data trusts. A well-functioning NDT will help secure long term democratic AI access for all and act as a counter weight to the disproportionate power of large tech companies.

- A robust system of transparency and public disclosure (for example, mandatory reporting of AI and ADM).
- An AI Safety Commissioner (as recommended by the Australian Human Rights Commission[28]).
- Utilisation of 'RegTech' tools by the Australian Government to monitor risk-critical automated systems (for example, algorithmic trading in the financial markets).
- Sandboxes and Innovation Offices/Labs to facilitate the experimental application of AI and ADM systems in a safe environment.

[28] https://tech.humanrights.gov.au/artificial-intelligence/ai-safety-commissioner.

- Development, adoption, and promotion of standards to enable consistency and encourage competitiveness among developers (such as AustCyber and Standards Australia have recommended for cyber security).
- Corporate governance guidance on how directors and officers can discharge their duties and obligations resulting from novel AI and ADM risk.
- Use of technical tooling for explainability and impact measurement, such as the Gradient Institute's 'AI Impact Control Panel'[29] within government departments, or the mandatory use of such tools (and publication of AI and ADM systems' calibration) by businesses active in Australia.
- Development of greater legal clarity surrounding liability for the outputs of AI and ADM systems, particularly in the context of (i) compensable harms to persons and (ii) government decision-making.

3.1 The possibility for a legislative approach

Mindful of the discussion in Section 2 above, Australia's position as a global leader in digital governance could be promoted by adopting a legislative framework. AI and ADM systems deployed in the public sector (i.e., by government agencies and their contractors) would, in our view, provide the logical first step for intervention, as demonstrated by some globally significant metropolitan initiatives. This might, for example, take the form of a model law promulgated at the federal level. Further consultation would be needed on the substantive principles and approaches to be adopted, but there is high-level consensus on some basic propositions that are, in our view, ready for operationalisation within a statutory framework. For example:

- Parties deploying an AI or ADM system must disclose (i) that they are doing so and (ii) some appropriate level of detail about the system in question and (iii) the humans involved in its operation.

[29] This tool assists decision-makers balance and constrain their system, prompting them to articulate their values and the trade-offs they are willing to make. https://gradientinstitute.org/ai-impact-control-panel/.

- AI and ADM systems are not 'agents' in the legal sense, and their 'action' (or 'behaviour'[30]) must be attributed to a human actor somewhere along the chain of creation, deployment, supervision, and control.[31]
- AI and ADM systems must be designed to keep humans in the loop in a meaningful sense (i.e., to apply human intelligence and potentially intervene in the automated process, not just as a scapegoat when things go wrong).

We would commend the Australian Government for leading the global effort to incorporate principles such as these into a binding legal framework. This would, in our view, certainly include a public AI register,[32] deploying transparency assessment methodology and ML observability tools,[33] and mandatory reporting and disclosure on certain matters (akin to financial services reporting). This approach could then be extended to other domains within the private sector in which procedural and substantive fairness are essential (e.g., credit scoring).

3.2 Facilitating the development of appropriate liability regimes for AI and ADM

Beyond textbook 'command and control' regulation, it is important to remember the inherent regulative function of private law liability regimes. These allow consumers and citizens to vindicate their rights and interests when impinged by the actions of another. For example, the law of torts imposes liability on categories of intentional, reckless, and negligent action that causes harm to others.[34] The problem currently is that many AI and ADM systems—especially data-driven systems that fall closer towards the 'intelligent' and 'autonomous' end of the spectrum—complicate notions of agency that underlie private law liability regimes and potentially confound their operation.[35]

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[30] See, eg, Mireille Hildebrandt, *Smart Technologies and the End(s) of the Law: Novel Entanglements of Law and Technology* (Edward Elgar 2015).
[31] In the private law context, for example, attribution is essential for the proper operation of fault-based compensation regimes; in the public law context, attribution is essential for the proper administration of accountability and liability doctrines.
[32] 'Public AI Registers', (White Paper, Saidot, September 2020) <https://algoritmeregister.amsterdam.nl/wp-content/uploads/White-Paper.pdf>.
[33] See, e.g., https://atarc.org/project/information-technology-artificial-intelligence-machine-learning-ml-model-transparency/.
[34] The 'regulative' function of private law is a staple of economic analysis in law; see, eg, Anthony Ogus, 'The Relationship Between Regulation and Tort Law: Goals and Strategies" in W.H. van Boom, Meinhard Lukas and Christa Kissling (eds.), *Tort and Regulatory Law* (Springer 2007), 377; Peter Cane, 'Tort Law as Regulation' (2002) 31 *Common Law World Review* 305.
[35] See J.G. Allen, 'Agency and Liability' in Charles Kerrigan (ed), *Artificial Intelligence: Law and Regulation* (Edward Elgar 2022); Julie-Ann Tarr, AA Tarr, and AJ George, 'Autonomous Vehicles: Regulatory, Insurance, and Liability Issues' (2021) 49 *Australian Business Law Review* 171.

We believe that the Australian Government could play a role, for example through the law reform process, in catalysing world-leading private law liability regimes for AI and ADM systems. In particular, it would be worthwhile to explore industry-specific (e.g., driverless vehicles) or general strict/no-fault liability regimes coupled with mandatory insurance (building on experience with the motor vehicle accident insurance).

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// 4. Conclusion

We are grateful to the Department of Prime Minister and Cabinet for conducting this consultation process, for the thoughtful and well-referenced Issues Paper, and for the opportunity to make this submission.

We welcome any further queries or discussion in respect of the whole or any part of this submission.

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